

PROPOSED Rep-YOLO11s NEURAL NETWORK ARCHITECTURE & RE-PARAMETERIZATION

CoordConv Spatial Priors • Multi-Scale CSPDarknet Backbone • BiFormer Dynamic Routing • Structural RepConv • Focal-EIoU Head

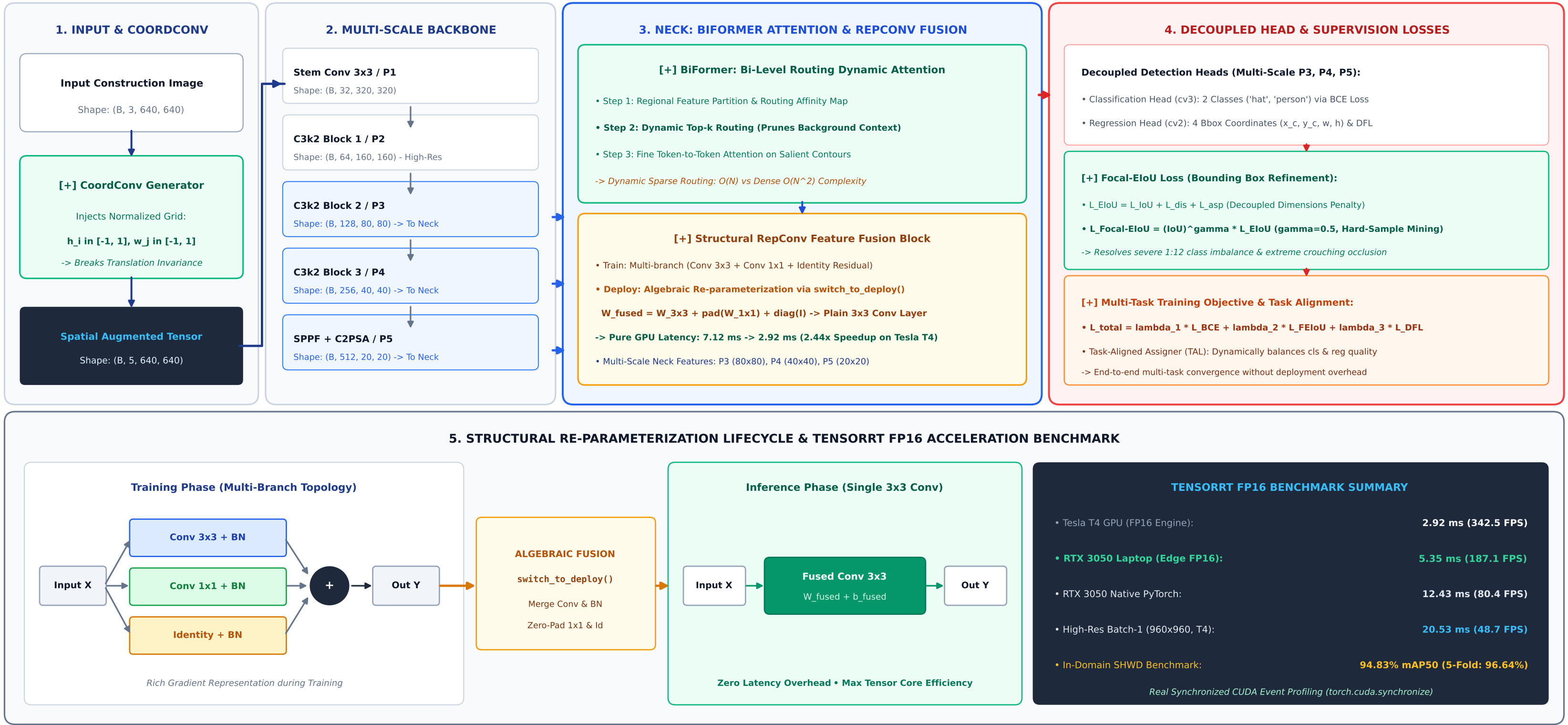


Figure 2: Complete Neural Network Architecture of Rep-YOLO11s displaying CoordConv coordinate generation, multi-scale CSPDarknet backbone, BiFormer sparse attention neck, structural re-parameterization lifecycle (training multi-branch to single-path inference), and decoupled heads with TensorRT FP16 acceleration.